National College of Ireland

Cloud Platform Programming

Object Storage Services

Outline

- Introduction to Cloud Storage Services
- Object Storage Services

Introduction to Cloud Storage Services

- Cloud storage enables us to save data and files in an off-site location
- Cloud storage services are
  - Cost-effective (pay per use)
  - Scalable
  - Elastic
- CSP responsible to maintain and secure the servers that host the data and related infrastructure, and provide access to data

Introduction to Cloud Storage Services

- Cloud Storage Types
  - Block
  - File
  - Object

Introduction to Cloud Storage Services

- Block Storage
  - Data is organized into large volumes named “blocks”
  - Each block represents a separate hard drive
  - Fast, efficient, and reliable data transportation

Introduction to Cloud Storage Services

- File Storage
  - Saves data in the hierarchical file and folder structure

Introduction to Cloud Storage Services

- Object Storage
  - Data is managed as objects
  - Each object includes the data in a file, its metadata, and an identifier
  - Objects are stored in repositories
  - Scalable solution to store large amounts of unstructured data

Cloud Object Storage – Advantages

- Durable
- Available
- Scalable
- Flexible
- Secure

Cloud Object Storage – Use Cases

- Disaster recovery/ Backup
- AI/ Analytics
  - Use object store as a data lake
- Cloud Native Workloads
- Data Archive

Cloud Object Storage Services – Examples

- Amazon Simple Storage Service (Amazon S3)
- Azure Blob Storage
- IBM Cloud Object Storage
- Google Cloud Storage

Amazon Simple Storage Service (Amazon S3)

- A cloud object storage service that provides
  - Industry-leading scalability
  - Data availability
  - Security
  - Performance
- Designed for 99.999999999% (11 9's) of data durability
  - Automatically creates and stores copies of all S3 objects across multiple systems

source: https://aws.amazon.com/s3

Amazon S3 – Use Cases

- Store and secure data for
  - Content storage and distribution
  - Backup and restore
  - Archive
  - Big data analytics, and data lakes
  - Static website hosting

source: https://aws.amazon.com/s3

Amazon S3 – Terminology



Image source: S3

- Amazon S3 stores objects in buckets
- A **bucket** is a container for objects stored in S3
  - When we create a bucket we choose the geographical region (AWS Region) where the bucket is created
  - We can control access for each bucket
    - Set access for creation, deletion, and listing of objects
    - View access log for the bucket and its objects

source: https://docs.aws.amazon.com/AmazonS3/latest/dev/Introduction.html

Amazon S3 – Terminology



Image source: S3

- Amazon S3 stores objects in buckets
- An **object** is the fundamental entity stored in S3
  - e.g. text files, photos, audio, video files, etc.
  - Consists of object data and any optional metadata
    - Metadata - a set of name-value pairs that describe the object
      - e.g. last date modified, HTTP metadata – Content-Type

source: https://docs.aws.amazon.com/AmazonS3/latest/dev/Introduction.html

Amazon S3 – Terminology



Image source: S3

- Amazon S3 stores objects in buckets
- A **key** is the unique identifier of an object within a bucket
- An object is uniquely identified within a bucket by a key and a version

source: https://docs.aws.amazon.com/AmazonS3/latest/dev/Introduction.html

Amazon S3 – Terminology



Image source: S3

- Amazon S3 stores objects in buckets
- AWS **Region** - when we create a bucket we choose the geographical region where the bucket is created
- Objects stored in a region are kept in that region, unless we explicitly transfer them to another region

source: https://docs.aws.amazon.com/AmazonS3/latest/dev/Introduction.html

Amazon S3 – Creating a Bucket

- The bucket name must be DNS-compliant
  - The name of an S3 bucket must be **unique** across all regions!
  - Be between 3 and 63 characters long
  - Bucket names can consist only of lowercase letters, numbers, dots (.), and hyphens (-).
  - Start with a lowercase letter or number
  - For best compatibility, avoid using dots (.) in bucket names
    - Except for buckets that are used only for static website hosting

Amazon S3 – Creating a Bucket

- A bucket is a flat container of objects
  - It does not provide any hierarchical organization (e.g. file system)
- We can create a logical hierarchy by using object key names (prefix) and delimiters that imply a folder structure
  - e.g. photo1.jpg
    - photos/2020/hiking/photo2.jpg
    - photos/2020/city-break/photo3.jpg

Amazon S3 – Uploading an Object

- Each Amazon S3 object has
  - Data
  - A key
  - Metadata
- You can set object metadata at the time you upload it
- The only way to modify object metadata after the object is uploaded, is to make a copy of the object and set the metadata

Amazon S3 – Uploading an Object

- Object key - unique identifier of an object within a bucket
- The name for a key is at most 1,024 bytes (UTF-8 encoding)
- The key name can contain
  - Alphanumeric characters [0-9, a-z, A-Z]
  - Special characters /, !, -, _, ., *, ', (,)

Amazon S3 – Uploading an Object

- Metadata - a set of name-value (key-value) pairs that describe the object by providing additional information
- System-defined metadata
  - e.g. object creation date, object size, object version, server-side encryption
- User-defined metadata
  - optional user-defined metadata names must be prefixed with "x-amz-meta-”
  - e.g. x-amz-meta-alt-name

Working with S3

Working with S3

- AWS Management Console
- S3 CLI
- REST API
- AWS SDKs

S3 API Reference - PutObject

- Amazon Simple Storage Service
  - Bucket and object-level operations
- AWS S3 Control
  - Account-level operations

source https://docs.aws.amazon.com/AmazonS3/latest/API/Type_API_Reference.html

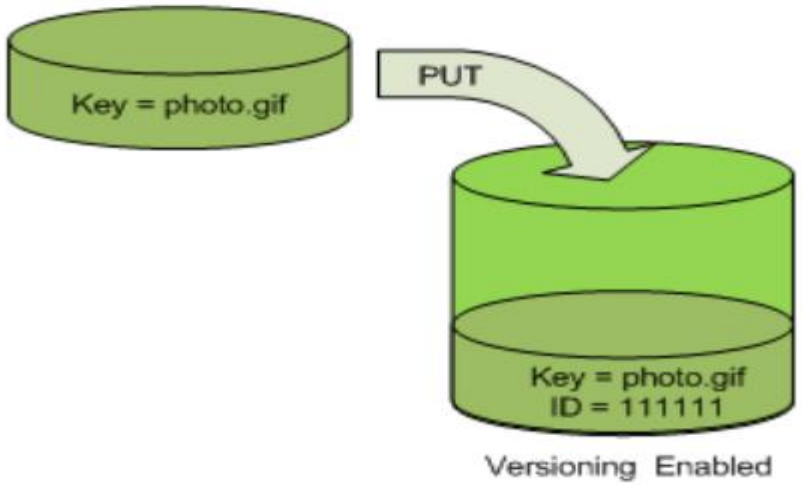
S3 API Reference - PutObject

- Adds an object to a bucket
- Can upload a single object of up to 5 GB
- For larger objects (up to 5 TB per object), use the multipart upload API
  - can use a multipart upload for objects from 5 MB to 5 TB
- If it receives multiple write requests for the same object simultaneously, it overwrites all but the last object written

source: https://docs.aws.amazon.com/AmazonS3/latest/API/API_PutObject.html

Adding Objects to a Versioning-Enabled Bucket

- If versioning is enabled on a bucket, Amazon S3 automatically adds a unique version ID to every object stored (using PUT, POST, or COPY) in the bucket



source: https://docs.aws.amazon.com/AmazonS3/latest/dev/AddingObjectstoVersioningEnabledBuckets.html

S3 API Reference - GetObject

- Retrieves objects from Amazon S3

source: https://docs.aws.amazon.com/AmazonS3/latest/API/API_GetObject.html

S3 API Reference - SelectObjectContent

- Filters the contents of an Amazon S3 object based on a SQL statement
- In the request is sent the SQL expression and the data serialization format (e.g. JSON, CSV, Apache Parquet)
- S3 uses this format to parse object data into records, and returns only records that match the SQL expression

source: https://docs.aws.amazon.com/AmazonS3/latest/API/API_SelectObjectContent.html

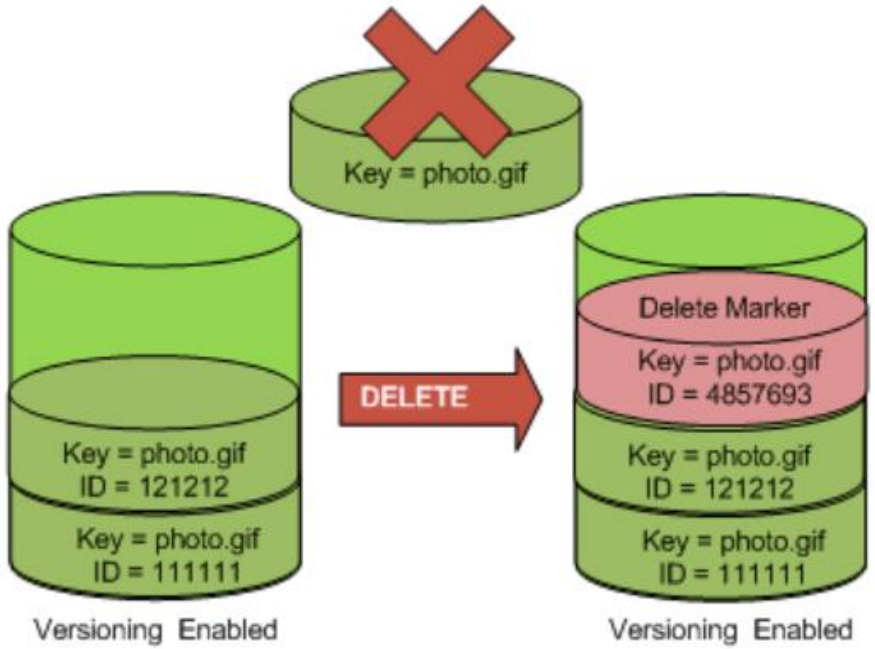
S3 API Reference – DeleteObject(s)

- DeleteObject operation - deletes an object from a bucket
- DeleteObjects operation - deletes multiple objects from a bucket using a single HTTP request

source https://docs.aws.amazon.com/AmazonS3/latest/API/API_DeleteObjects.html

Deleting objects from a Versioning-Enabled Bucket

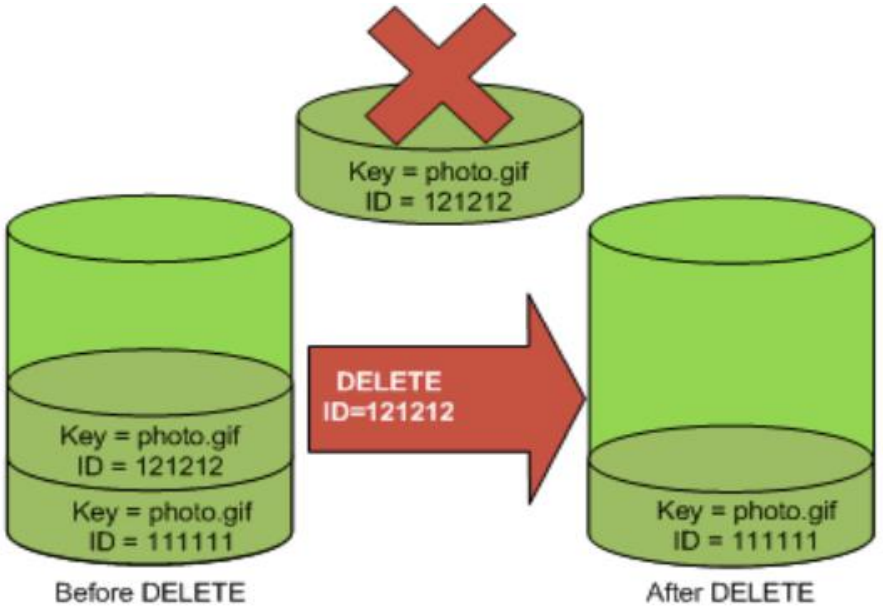
- When bucket versioning is enabled, a simple DELETE cannot permanently delete an object



source: https://docs.aws.amazon.com/AmazonS3/latest/dev/DeletingObjectVersions.html

Deleting objects from a Versioning-Enabled Bucket

- When bucket versioning is enabled, to permanently delete an object the object versionID must be specified



source: https://docs.aws.amazon.com/AmazonS3/latest/dev/DeletingObjectVersions.html

Resources

- Amazon Simple Storage Service (Amazon S3) - https://aws.amazon.com/s3/
- Azure Blob Storage - https://azure.microsoft.com/en-us/services/storage/blobs/
- IBM Cloud Object Storage - https://www.ibm.com/cloud/object-storage
- Google Cloud Storage - https://cloud.google.com/storage

Resources

- Amazon S3 API Reference: https://docs.aws.amazon.com/AmazonS3/latest/API/Type_API_Reference.html